

MALILLA, Sweden

WMO: 025660

Lat: 57.3845N

Lon: 15.8009E

Elev: 98

StdP: 100.15

Time Zone: 1.00 (EUC)

Period: 86-07

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-18.8	-14.5	-20.8	0.6	-17.5	-16.7	0.9	-13.0	11.2	-2.4	10.0	4.3	0.6	340	0.566

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.0	28.0	17.0	25.9	16.4	23.9	15.5	18.5	25.0	17.5	23.4	16.5	21.8	2.9	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
16.5	11.9	19.5	15.3	11.0	18.7	14.4	10.3	18.0	52.7	24.2	49.6	23.9	46.5	21.6	22.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.4	6.7	5.3	DB	-22.2	30.4	4.4	2.6	-25.4	32.2	-28.0	33.7	-30.5	35.1	-33.7	37.0	(4)
(5)			WB	-22.4	20.0	4.4	1.4	-25.6	21.0	-28.2	21.8	-30.7	22.6	-33.8	23.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	6.5	-1.7	-1.4	0.9	5.3	10.1	13.6	16.3	15.3	11.0	6.6	1.9	-0.8	(6)	
	DBStd	7.59	5.52	5.42	4.50	3.86	3.25	3.00	2.96	3.01	2.99	3.81	3.92	5.02	(7)	
	HDD10.0	1877	364	320	284	148	39	4	0	1	24	113	243	336	(8)	
	HDD18.3	4357	622	553	542	390	256	143	78	104	219	363	492	594	(9)	
	CDD10.0	587	0	0	0	8	42	113	196	164	55	8	1	0	(10)	
	CDD18.3	26	0	0	0	0	0	2	15	8	0	0	0	0	(11)	
	CDH23.3	541	0	0	0	4	23	87	291	132	3	0	0	0	(12)	
	CDH26.7	123	0	0	0	0	1	14	86	23	0	0	0	0	(13)	
(14) Wind	WSAvg	1.8	2.1	2.1	1.9	1.8	1.8	1.9	2.1	1.6	1.6	1.5	1.5	1.8	(14)	
(15) Precipitation	PrecAvg	604	35	33	24	32	50	58	89	66	57	54	59	45	(15)	
	PrecMax	773	61	71	49	75	99	127	224	140	166	130	170	84	(16)	
	PrecMin	475	2	5	1	3	11	0	20	7	5	14	14	8	(17)	
	PrecStd	68	17	17	12	20	20	32	45	31	44	29	31	20	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.1	10.4	14.9	23.1	25.8	28.2	31.5	29.0	23.8	16.8	12.0	9.8	(19)	
		MCWB	7.3	8.0	8.0	13.6	15.3	16.9	18.9	17.8	15.1	12.9	9.7	8.3	(20)	
	2%	DB	7.3	8.3	11.5	18.5	23.4	25.6	29.2	26.9	20.9	14.7	10.3	8.0	(21)	
		MCWB	5.8	6.1	6.2	10.8	14.3	15.9	17.9	17.2	14.5	12.0	8.6	6.8	(22)	
	5%	DB	5.8	6.9	9.0	15.5	21.1	23.6	26.7	24.7	18.6	13.4	8.9	6.8	(23)	
		MCWB	4.3	5.0	5.4	8.9	12.7	14.9	16.6	16.6	13.5	11.2	7.5	5.6	(24)	
	10%	DB	4.5	5.6	7.1	13.0	18.6	21.6	24.2	22.5	16.8	12.2	7.7	5.5	(25)	
		MCWB	3.1	3.8	4.5	7.7	11.3	14.1	15.7	15.8	12.6	10.3	6.6	4.5	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.6	8.4	8.7	13.9	16.3	18.4	19.7	20.2	16.7	14.2	10.4	8.4	(27)	
		MCDB	8.9	10.0	13.0	22.5	23.6	25.1	28.7	25.6	20.5	15.9	11.5	9.3	(28)	
	2%	WB	6.0	6.5	7.1	11.5	14.8	17.0	18.6	18.6	15.4	12.8	8.8	6.8	(29)	
		MCDB	7.1	8.0	10.1	17.2	21.7	23.4	26.6	23.6	18.8	14.0	9.8	7.7	(30)	
	5%	WB	4.6	5.4	5.9	9.6	13.3	15.9	17.5	17.7	14.4	11.7	7.8	5.8	(31)	
		MCDB	5.6	6.7	8.3	14.4	19.5	21.8	24.4	22.5	17.3	12.9	8.7	6.7	(32)	
	10%	WB	3.3	4.0	4.7	8.2	12.1	14.8	16.4	16.6	13.5	10.6	6.8	4.5	(33)	
		MCDB	4.2	5.4	7.0	12.2	17.3	20.2	22.1	21.1	15.9	11.8	7.6	5.3	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	5.3	6.9	8.8	11.2	12.8	11.8	12.0	11.1	9.9	7.8	5.4	5.3	(35)	
		MCDBR	5.1	6.7	12.3	16.5	18.9	17.0	17.7	15.4	13.5	9.3	6.1	5.6	(36)	
	5% WB	MCWBR	4.7	5.2	7.7	9.3	10.1	8.1	7.8	6.9	7.4	6.8	5.2	5.1	(37)	
		MCWBR	4.9	5.7	10.7	15.0	16.4	14.3	14.9	12.9	10.9	7.8	5.6	5.4	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.291	0.323	0.340	0.377	0.375	0.364	0.390	0.384	0.353	0.342	0.318	0.280	(40)	
		taud	2.286	2.324	2.357	2.326	2.391	2.444	2.388	2.403	2.478	2.426	2.329	2.233	(41)	
	Ebn at Noon		580	721	813	835	862	876	841	820	796	693	538	474	(42)	
		Edh at Noon	48	72	90	108	108	104	108	100	80	65	47	38	(43)	
(44) All-Sky Solar Radiation	RadAvg		0.37	0.98	2.47	4.04	5.27	5.70	5.16	4.17	2.84	1.30	0.48	0.27	(44)	
	RadStd		0.07	0.19	0.26	0.52	0.48	0.30	0.60	0.40	0.37	0.23	0.09	0.03	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (49 neighbors)	+0.40	N/A	N/A	N/A	N/A	N/A	-123	-148	N/A	N/A

Nomenclature: See separate page